CLASS REFERENCES UNDER TRANSFER

Graph context constructs the classifier

Contribution: support processing can alter discrimination without changing query vectors or attention routing. A tested value-path explanation identifies one transfer failure; the public benchmark shows why suppressing context is not a general remedy. The remaining question is what predicts that boundary.

A Support-context utility varies across sources

Suppression minus intact, pooled AUC points; two-stream mean

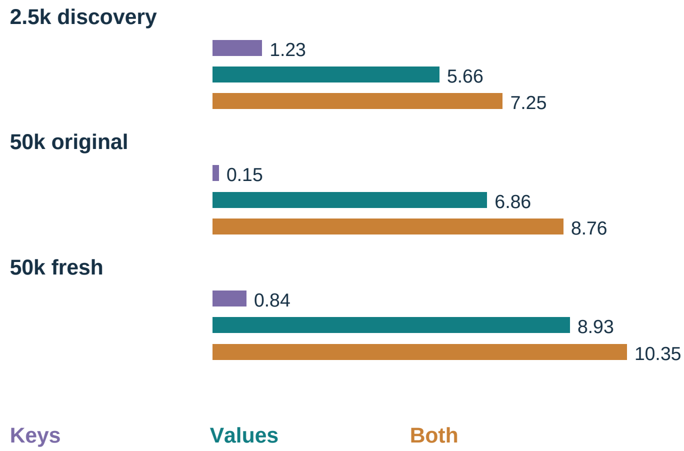
	Pol.	Elect.	Pages	Bots	Susp.
COVID	-0.1*	+0.0*	+0.1	-4.0	+1.2
Political	-5.7	-0.4	-3.9	-2.4	-2.8
Hong Kong	+9.9	+2.0	+2.5	-7.5	+0.6*
Election	-0.2	-0.0*	-1.3	-10.1	+0.2*
Pages	+1.8	+1.3	+0.9	-1.6	+1.4
Midterm	+3.1	+0.3	+0.8	-0.6	-0.1*
Bots	+0.8	+0.1	+0.1*	-2.6	+0.2*
Ukraine	-0.1	-0.0	-0.0*	-5.4	+1.7
Suspended	+7.4	+0.1*	-0.6	-8.3	+1.4

Teal: suppression helps. Red: it hurts. *: stream signs differ.

Hong Kong is an extreme, not the only affected source. This map establishes heterogeneity, not a validated predictor of its sign.

B Values carry more of the effect than routing

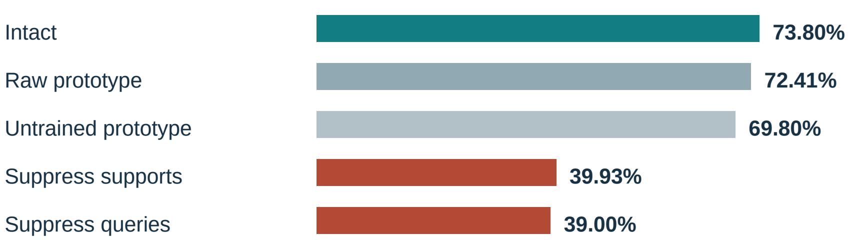
Hong Kong to political; within-episode AUC change (points)



Value-only substitution preserves attention and query vectors exactly. The value-over-key ordering transfers to both 50k streams; baseline/removal endpoints were already known. This is one source-target pair, not cross-source prediction.

C The strongest counterexample: a useful public model needs both roles

Original-style Wiki to FB15K-237, fixed 8,001-update checkpoint; 500 paired 20-way episodes



Both scientific intervals exclude zero:
Support suppression: -33.87 points
95% CI [-34.55, -33.19]
Query suppression: -34.80 points
95% CI [-35.48, -34.14]
The role-conflict prediction fails.

Native exceeds raw by 1.38 points [0.74, 2.03] and untrained by 3.99 [3.36, 4.62]. Intervals are paired episode-bootstrap 95% CIs conditional on this graph/checkpoint, not independent training seeds. Public training-mode BatchNorm couples episodes' support/query representations; this is not fixed-query mediation.

Explanation supported

In the social failure, context-altered support **values** construct a poor class reference even with queries and routing held fixed. Suppression compresses the support-driven contrast while a class preference survives. At 50k, political AUC rises 56.37% to 65.92%, but accuracy falls 49.19% to 25.15%: **better ranking is not classifier repair.**

Exact contribution and missing link

A causal localization and a tested value-path prediction across training configurations, plus a concrete warning about evaluating class-reference geometry using AUC alone. **No general sign predictor or deployable method is established.** The public result rules out universal suppression; it does not yet explain which learned operating conditions cause the failure.

Verification disclosure: GPU repeats differ numerically. The original logit-tolerance check failed; a dated amendment before baseline/role outcomes required exact labels and all 40,000 predicted classes, which matched. No tolerance was enlarged. Native architecture, sum aggregation, released split/BN behavior and scientific criteria were retained.